

OPTIMUS PRIME / G1 EXPEDITION

THE MOUNTAIN AMPLIFIES FAILURE

A SLIP BECOMES A FALL.

A FALL BECOMES A RESCUE.

G1 must use tools, recover, and know when to stop.



SIMPLIFIED

OPTIMUS PRIME / G1 EXPEDITION

A SLIP BECOMES A RESCUE.

G1 MUST RECOVER BEFORE A HUMAN MUST INTERVENE.

Extreme terrain turns small failures into high-consequence missions.



FOUR SKILLS FOR EXTREME TERRAIN.

ONE EXPEDITION SYSTEM



01 SELF-ARREST

STOP THE
SLIDE



02 FALL RECOVERY

GET BACK
UP



03 CONTROLLED RAPPEL

DESCEND WITH
CONTROL



04 TEAM LIFT

MOVE HEAVIER
LOADS TOGETHER

FOUR SKILLS FOR EXTREME TERRAIN.

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HOW SINGLE-ROBOT SKILLS LEARN

MUJOCO / PPO / HIERARCHICAL CONTROL

SELF-ARREST PPO

125-D OBSERVATION →

2 × 256 TANH →

14 ARM RESIDUALS

100 Hz · ≈200k actor+critic params

FIXED-LINE RECOVERY

FALL + ROPE STATE →

2 × 256 TANH →

4-ACTION HANDOFF

50 Hz · pretrained 29-DoF WBC prior

REWARD SHAPING / PHYSICS MUST EXPLAIN SUCCESS

01 TOOL CONTACT

rigid pick contact
blade 22–42° · snow load

02 BODY CONTROL

chest-down pose · grip
wrists stay ahead of torso

03 TERMINAL GATES

no passive stop
stable stand earns success

CURRICULUM

0.25–0.75 m/s

START EASY

→ 5.0 m/s

SPEED

→ ±40° / ±1.5 m/s

HEADING + CROSS-SLOPE

→ HARD ANCHORS

ROLL + ADVERSARIAL FALLS

→ 9/9 + 60/60

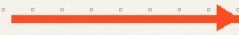
FROZEN SELECTION

ONE POLICY. MANY FALLS.

SINGLE-ROBOT PPO

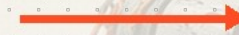
125 OBSERVATIONS

pose · speed · tool load



PPO 2×256

compact learned policy



14 ARM COMMANDS

plant and load the axe

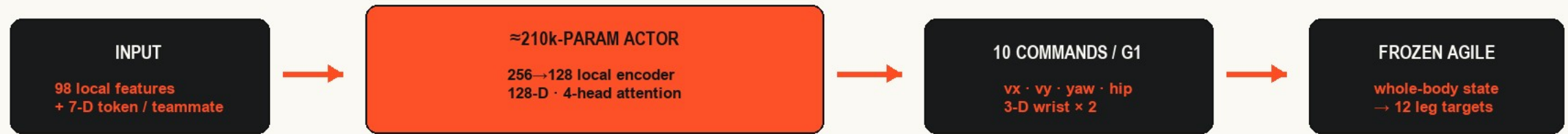
EASY SLIDES → 5 m/s + CROSS-SLOPE FALLS

SUCCESS COUNTS ONLY WHEN THE AXE PHYSICALLY BITES.

HOW THE TEAM POLICY LEARNS

ISAAC LAB / SHARED MAPPO / CENTRALIZED TRAINING

SHARED ATTENTION ACTOR / SAME WEIGHTS ON EVERY G1



TRAINING-ONLY CENTRAL CRITIC

TEAM STATE + PAYLOAD MASS

768 → 512 → 256

ACTOR: LOCAL + TOKENS ONLY
AT INFERENCE

MAPPO: 24-step rollouts · 5 epochs · $\gamma .99$ · GAE .95 · clip .2 · lr 3e-4

SHARED TEAM REWARD

+ TRACK

+ LEVEL

+ HEADING

+ LOAD BALANCE

+ UPRIGHT

+ LIFT PROGRESS

- EXTENSION / RATE / FALL

CURRICULUM

LIFT + LEVEL

0-24k steps

ADD CARRY + TURN

24k→150k

MASS 8→18 kg

by 180k

2 / 3 / 5 G1

shared actor

ONE SHARED POLICY. EVERY ROBOT.

COOPERATIVE MAPPO



LOCAL STATE

+ TEAMMATE TOKENS



SHARED MAPPO

10 COMMANDS / G1

Same actor weights on every robot.

LIFT FIRST → CARRY + TURN → HEAVIER LOADS

LIVEKIT CONNECTS THE EXPEDITION

ONE ROOM / TWO EXPLICIT CONTROL PLANES

ROBOT STATE PLANE / LIVEKIT DATA TRACKS



OPERATOR VOICE PLANE / LIVEKIT AGENTS + INFERENCE



VOICE SUPERVISES · LOCAL POLICIES CONTROL MOTION

LIVEKIT IS THE EXPEDITION BUS.

STATE + VOICE

STATE

ONE STATE STREAM / G1



LIVEKIT ROOM



EVERY ACTOR STAYS IN SYNC

VOICE

"MOVE THE LOG"



CONFIRM



GATED START INTENT

VOICE SUPERVISES. LOCAL POLICIES CONTROL MOTION.

WHAT WE MEASURED

FIXED TESTS / PHYSICAL OUTCOMES



SELF-ARREST

9 / 9 SCENARIOS

60 / 60 unseen falls arrested



FALL RECOVERY

RECOVERS TO STAND

then continues uphill



RAPPEL

2.00 m DESCENT

7 / 10 randomized starts passed



TEAM LIFT

50 / 50 SHARE

overload or imbalance → abort

FOUR SKILLS. MEASURED RESULTS.

FROZEN EVALUATION

ARRESTS PASSED

69 / 69

9 named + 60 unseen

ICY FIXED LINE

STANDS AGAIN

recover + continue

RAPPEL DESCENT

2.00 m

controlled braking

TARGET LOAD SPLIT

50% / 50%

balanced teamwork

STOP · RECOVER · DESCEND · COORDINATE